Data Analysis & Visualisation

Week 2

Raoul Grouls, 24-2-2026

Recap leerdoelen les 1

- de gestalt principes & five guidelines toepassen op visualisaties
- een virtual environment activeren met uv
- nieuwe features extraheren met behulp van regular expressions
- Een script vanaf de terminal opstarten
- click gebruiken voor command line arguments bij een script
- begrijpt de opzet van een project (src folder, data/raw en data/processed, pyproject.toml, notebooks) en kan dit zelf opzetten
- kan een eigen git-repo maken

- Regular expressions toepassen:
  - start ^
  - end $
  - or: [Bb]
  - ranges [a-zA-Z]
  - any char .
  - zero or more a*
  - one or more a+
  - not in range [^a-z]
  - shortcuts (\w, \s, \d)
  - lookbehind (?<=...)
  - lookahead (?=...)

leerdoelen les 2

- Leren toepassen van visualisatie principes
- Omgaan met venv, uv, path, scripts en git
- Oefenen met nieuwe features extraheren met behulp van regular expressions
- Vergelijken van categorieen met behulp van data visualisaties:
  - Barplots
  - Barbell plot
  - heatmaps
- Werken met palettes (en list comprehensions)
- Pandas
  - Pandas groupby & aggregate
  - Pandas cut

rollover for more detail

What Makes a Good Visualization?

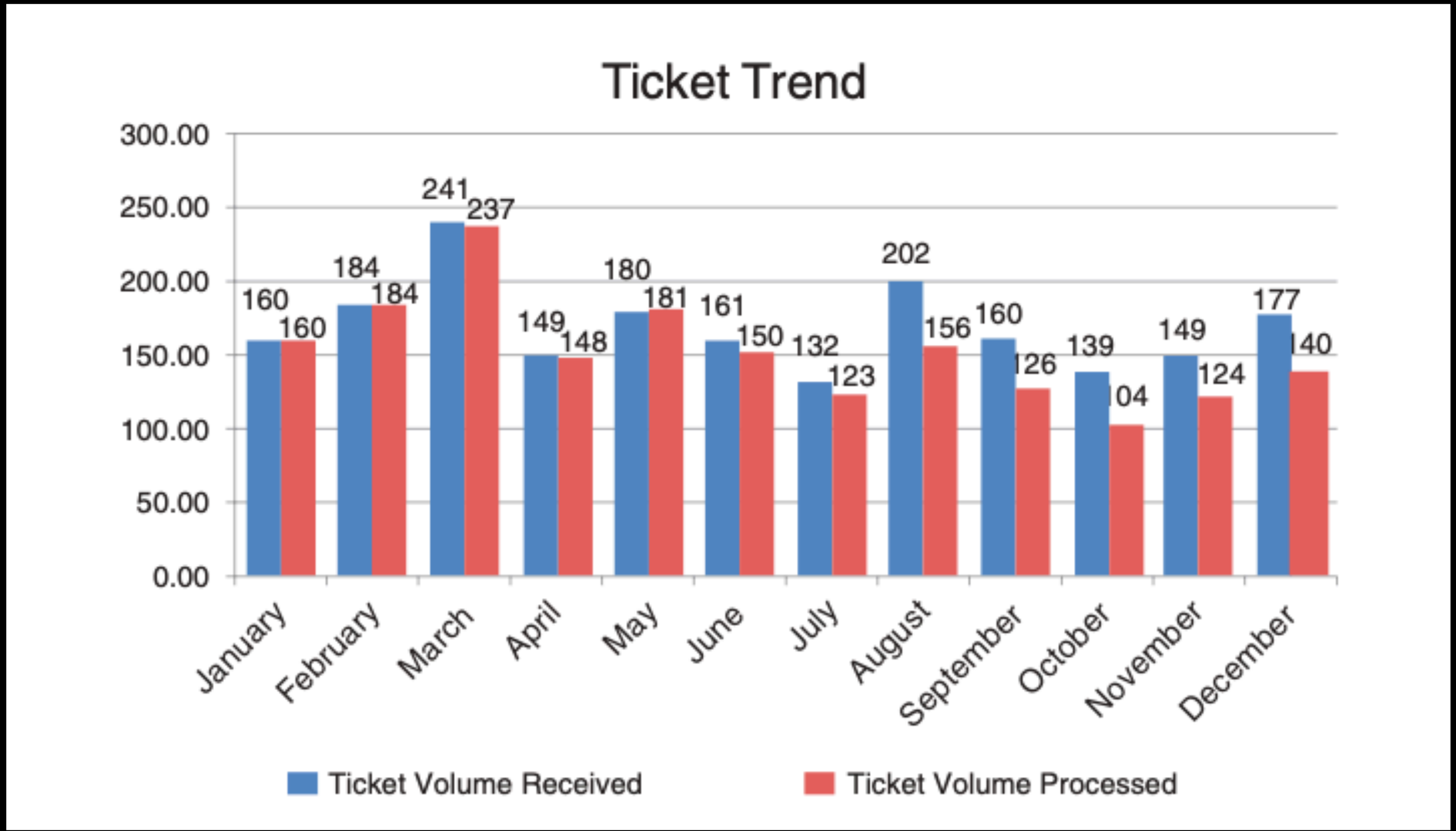
explicit (implicit)



David McCandless
InformationisBeautiful.net

taken from new book
Knowledge is Beautiful

find out more
bit.ly/KIB_Books


Ticket Trend
300.00
250.00
200.00
150.00
100.00
50.00
0.00
January
February
March
April
May
June
July
August
September
October
November
December
160
160
184
184
241
237
149
148
180
181
161
150
132
123
202
156
160
126
139
104
149
124
177
140
Ticket Volume Received
Ticket Volume Processed

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

Ticket volume over time



Data source: XYZ Dashboard, as of 12/31/2014 | A detailed analysis on tickets processed per person and time to resolve issues was undertaken to inform this request and can be provided if needed.

Survey Results

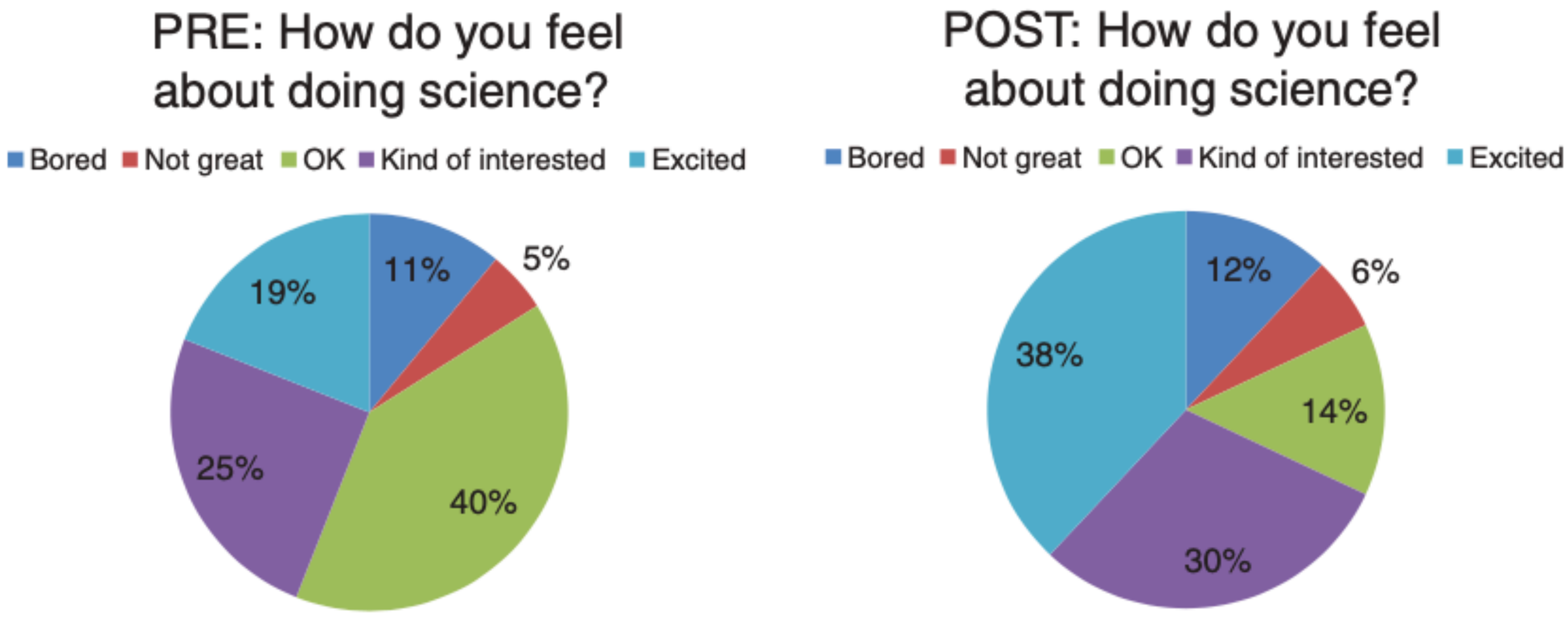


FIGURE 0.4 Example 2 (before): showing data

Pilot program was a success

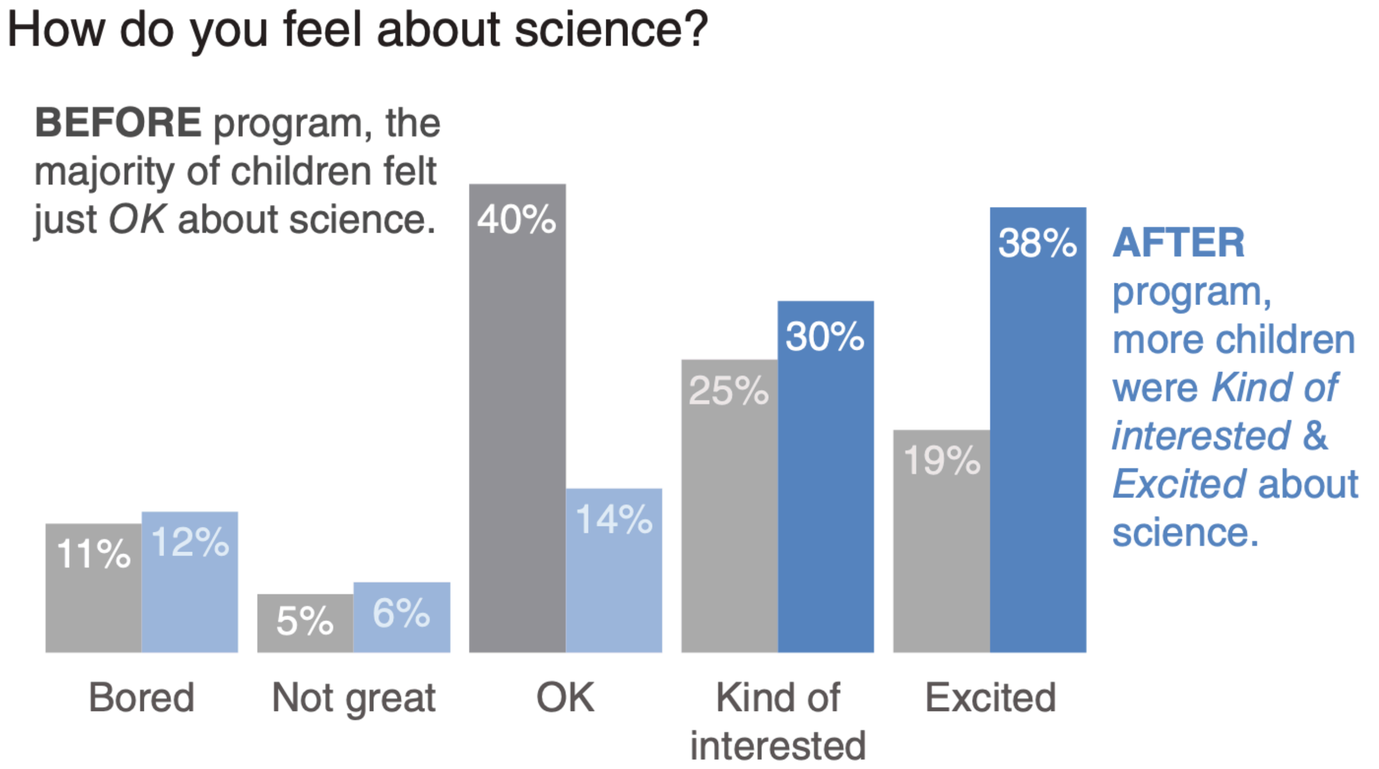


FIGURE 0.5 Example 2 (after): storytelling with data

Things to avoid

Things to avoid I

Ordering counts

- Problem: **Ordering counts** will ALWAYS give a top-k graph. But the fact that you can order some thing you can count doesnt necessarily MEAN something.
- Solution: think about what the **story** is you want to tell.

Things to avoid II

Spurious correlations

- Problem: You see a pattern and make conclusions because it seems to fit a story. But you ignore that this could be a statistical coincidence. Typical example is making conclusions about males and females, based on a chat with 5 people.
- Solution: Make sure you understand the weakness of your statistics. Eg distortions by applying normal-distributions to log-normal data, or making conclusions on less than 30 samples. Spot possible spurious correlations, and give alternative hypothesis for the pattern (eg, character instead of gender)

Things to avoid III

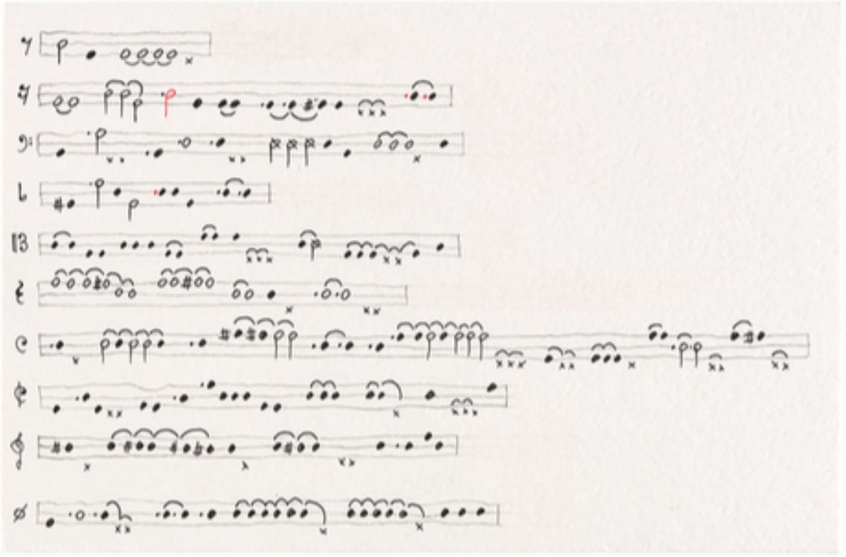
Lack of focus

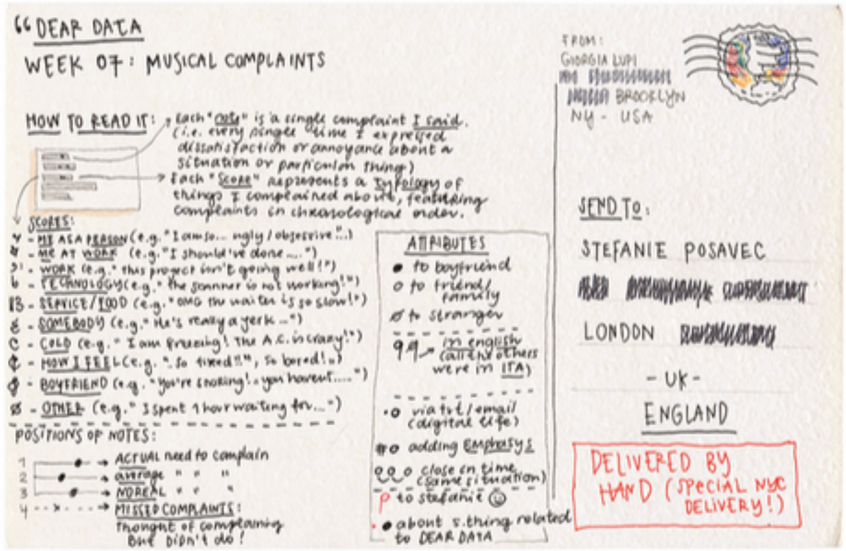
- Problem: You are busy showing all the data, exploring the data and now you are showing the full complexity of 5 different, uncorrelated patterns in a single plot. But you don't want to simplify because it took you 5 hours to create the plot.
- Solution: kill your darlings. Try to focus on the most compelling pattern you see and start again with a new plot.

Trade offs

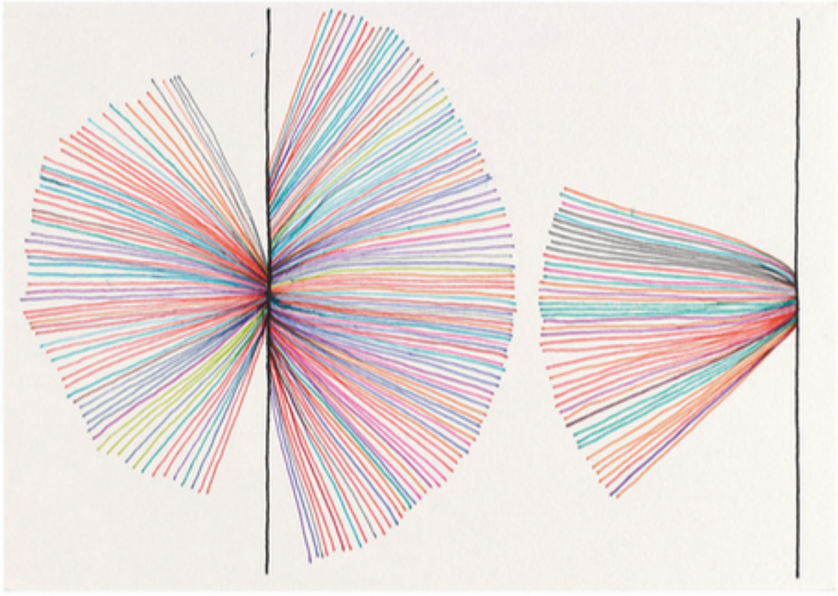
Afweging 1: snelheid

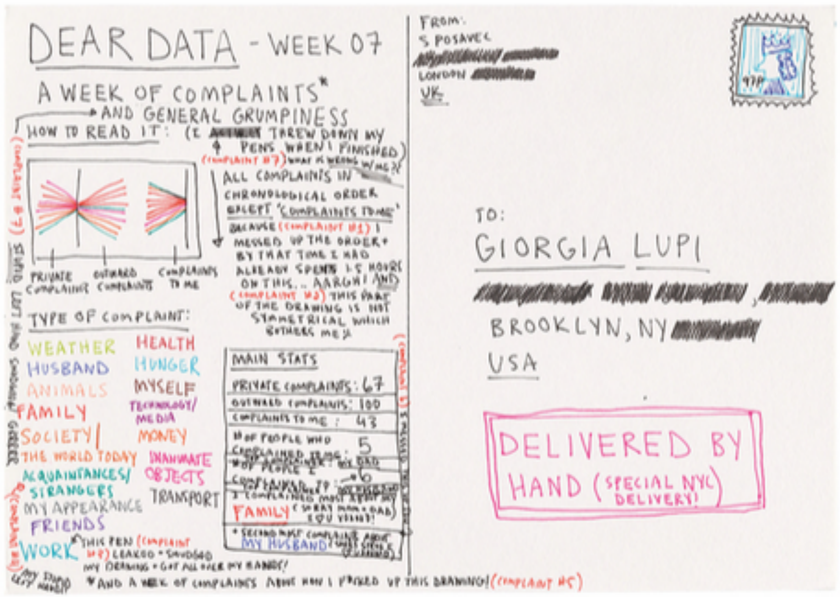
- Meer story, minder information?
- preattentive processing vs bestuderen?





What better visual reference than a musical score to show the repetitiveness of Giorgia's protests and the "level" of complaint: whether they are justified or totally out of place.





Note the hand-drawn stamps: these postcards were delivered in person in New York!

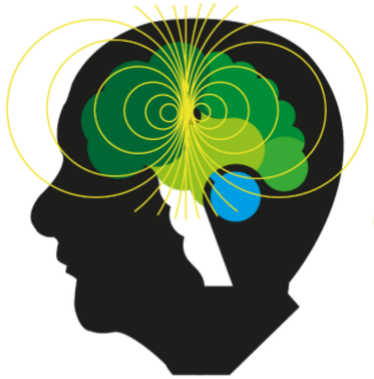
What is Consciousness?

Make up your own mind



A field that exists in its own parallel "realm" of existence outside reality so can't be seen.
(Substance Dualism)

A sensation that "grows" inevitably out of complicated brain states.
(Emergent Dualism)



A physical property of all matter, like electromagnetism, just not one the scientists know about.
(Property Dualism)



All matter has a psychic part. Consciousness is just the psychic part of our brain.
(Pan Psychism)



Simply, mental states are physical events that we can see in brain scans.
(Identity Theory)



Consciousness and its states (belief, desire, pain) are simply functions the brain performs.
(Functionalism)



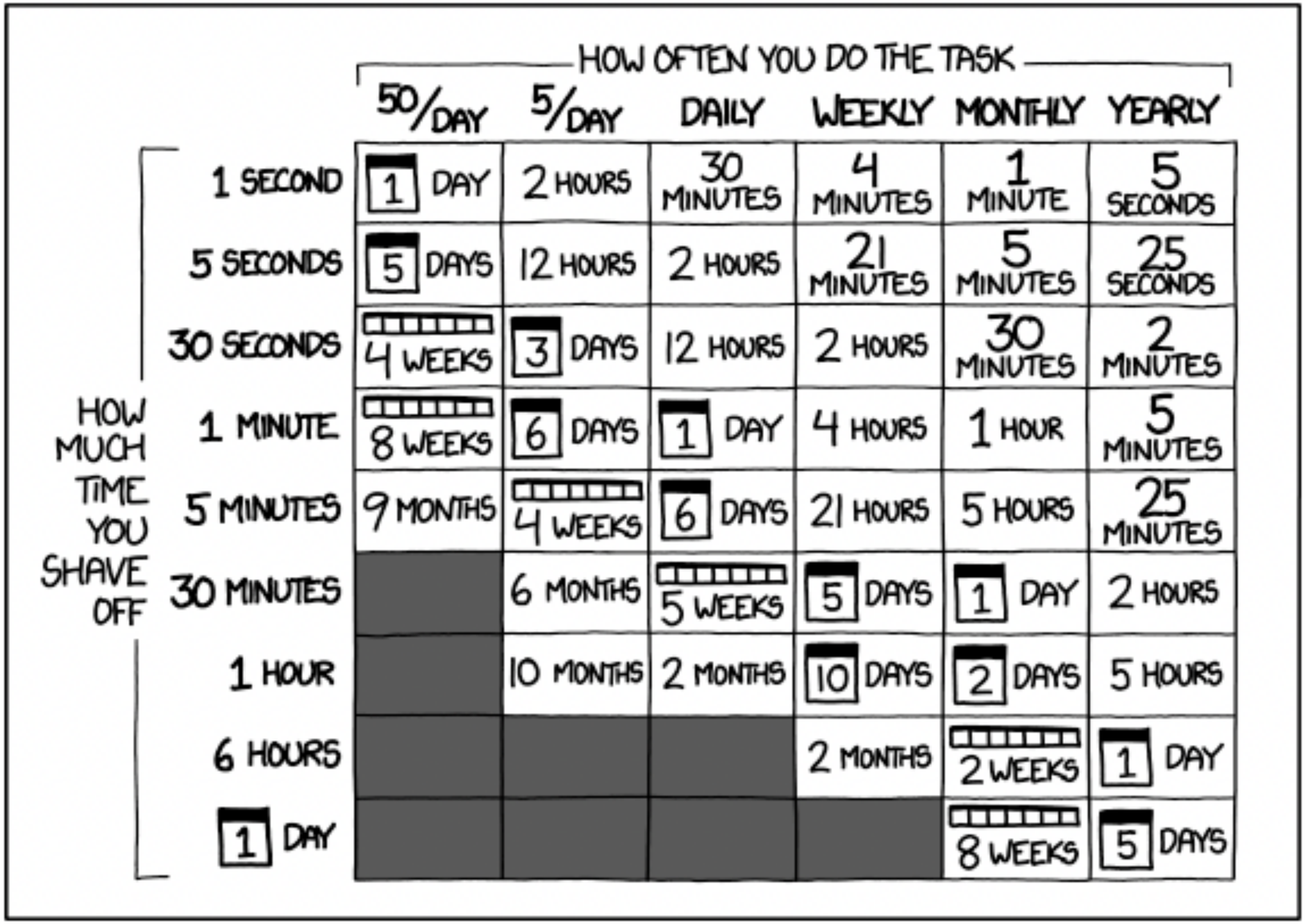
Literally just behaviour. When we behave in a certain way, we appear conscious.
(Behaviourism)



An accidental side-effect of complex physical processes in the brain.
(Epiphenomenalism)

HOW LONG CAN YOU WORK ON MAKING A ROUTINE TASK MORE EFFICIENT BEFORE YOU'RE SPENDING MORE TIME THAN YOU SAVE?

(ACROSS FIVE YEARS)



Afweging 2: confidence

- Absolute vs relative numbers
- Confidence comes with higher numbers

Errorbars

- Standard error
- Confidence intervals
- Standard deviation
- Range

Errorbars

Suppose statistically independent n samples $x_1, x_2, \ldots, x_n$ from a population

Suppose the population has a standard deviation of σ

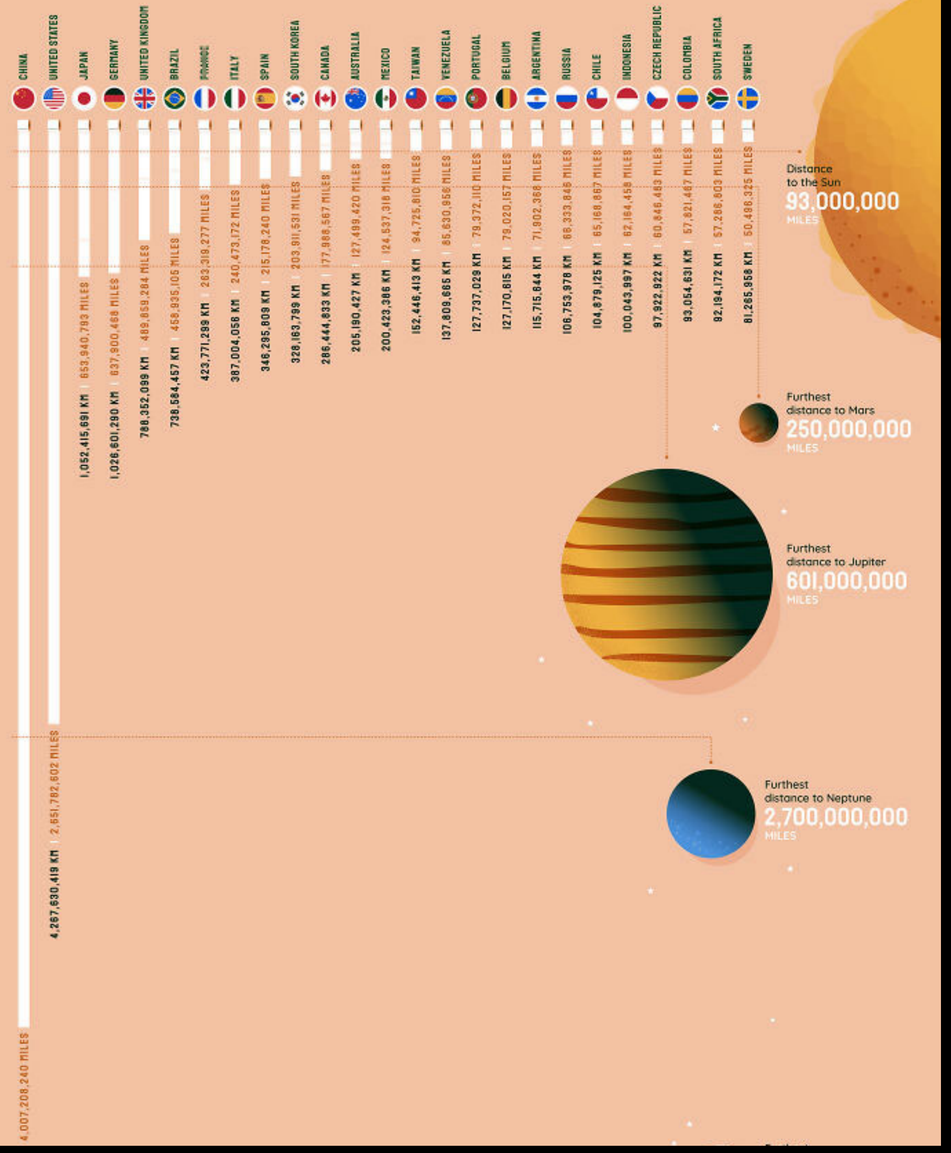
The standard error is calculated with $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Because we rarely know the σ of the population, we often estimate it with the sample standard deviation.

THE COUNTRIES THAT USE THE MOST TOILET PAPER

If you laid out all of the rolls of toilet paper used in each country in one year, what would that look like? For China it would be an incredible 4 billion miles long, which is further than the distance from Earth to Neptune. For the USA it's 2.65 billion miles, while both Japan and Germany use enough toilet paper to stretch out beyond Jupiter.

▸ TOILET PAPER USAGE PER YEAR PER COUNTRY



DEATH TOLL

[HIGHEST TO LOWEST]

the disease is new to medicine, and data is still coming in.

200M
Black Death (Bubonic Plague)
1347-1351



The plague originated in rats and spread to humans via infected fleas.

The outbreak wiped out 30-50% of Europe's population. It took more than 200 years for the continent's population to recover.

56M
Small Pox
1520



Smallpox killed an estimated 90% of Native Americans. In Europe during the 1800s, an estimated 400,000 people were being killed by smallpox annually. The first ever vaccine was created to ward off smallpox.

40-50M
Spanish Flu
1918-1919



30-50M
Plague of Justinian
541-542



The death toll of this plague is still under debate as new evidence is uncovered, but many think it may have helped hasten the fall of the Roman Empire.



25-35M
HIV/AIDS
1981-PRESENT



12M
The Third Plague
1855



5M
Antonine Plague
165-180



3M
17th Century Great Plagues
1665



1.1M
Asian Flu
1957-1958



1M
Russian Flu
1889-1890



1M
Hong Kong Flu
1968-1970



1M
Cholera 6 outbreak
1817-1923

A series of **Cholera** outbreaks spread around the world in the 1800s killing millions of people. There is no solid consensus on death tolls.



1M
Japanese Smallpox Epidemic
735-737



600K
18th Century Great Plagues
1817-1923



200K
Swine Flu
2009-2010



100-150K
Yellow Fever
LATE 1800s



11.3K
Ebola
2014-2016

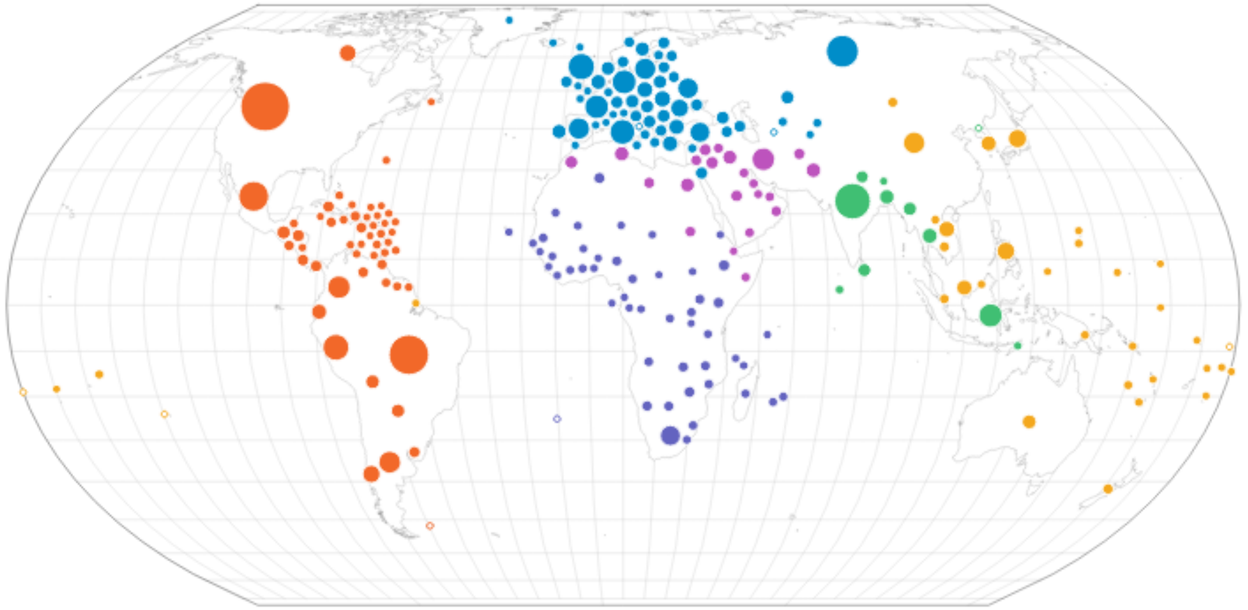
4.7K*
Novel Coronavirus (COVID-19)
2019-PRESENT

850
MERS
2015-PRESENT

770
SARS
2002-2003

Number of COVID-19 deaths reported to WHO (cumulative total)

World



WHO Regions Africa Americas Eastern Mediterranean Europe South-East Asia Western Pacific

7,031,216

Reported COVID-19 deaths

11 February 2024

Number of COVID-19 deaths reported to WHO (cumulative total)

World

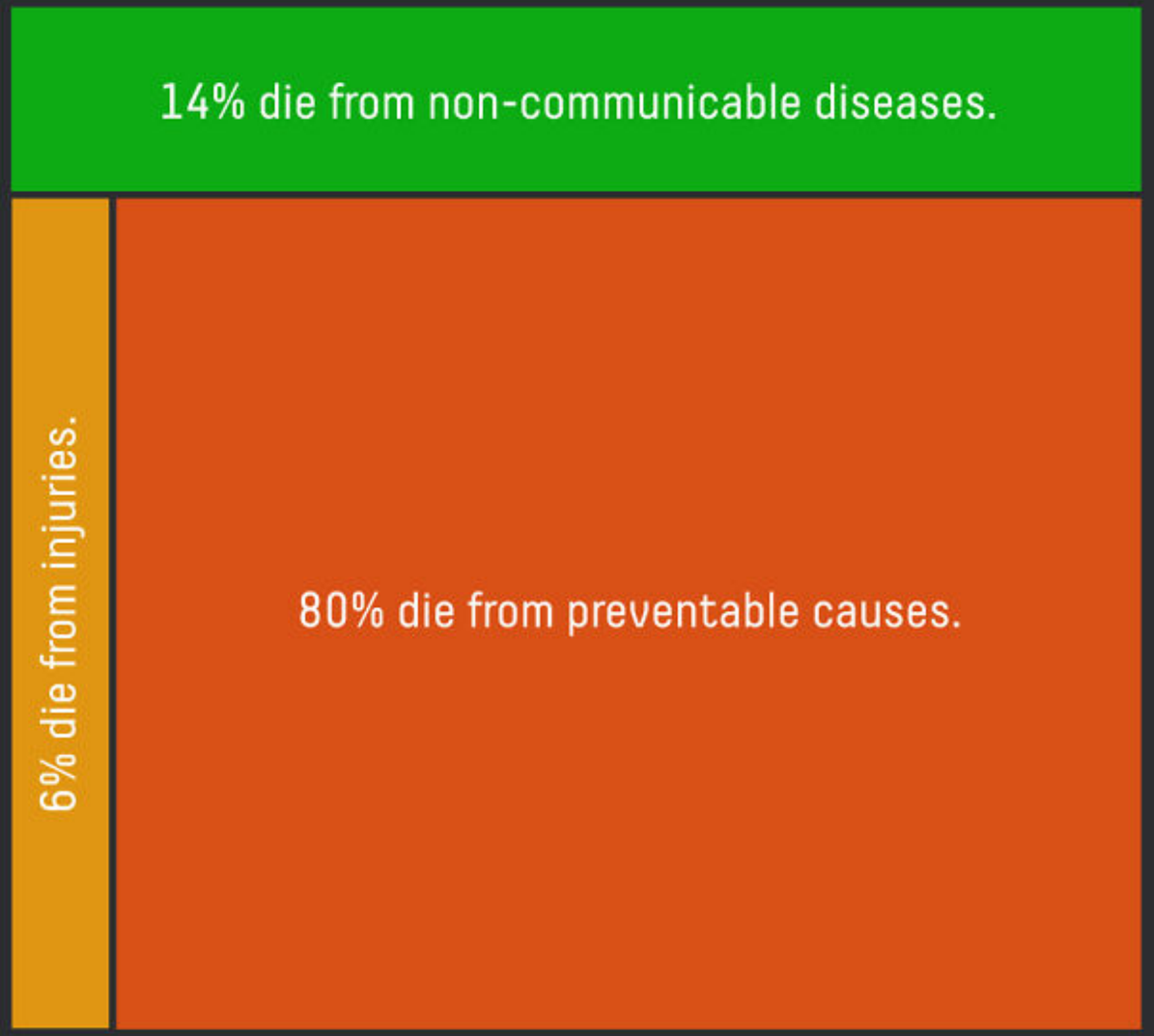
Country	Deaths
United States of America	1.2m
Brazil	702.1k
India	533.5k
Russian Federation	402.1k
Mexico	335k
United Kingdom of Great Britain And	232.1k

Show less

https://data.who.int/dashboards/covid19/deaths?n=c

Causes of death in children under 5

(2013)



Source: IHME

gatesnotes

Start with grey

Color palettes

https://seaborn.pydata.org/tutorial/color_palettes.html

- **Qualitative**: representing categorical data
- **Sequential**: perceptually uniform
- **Diverging**: both large low and high values are interesting and span a midpoint value







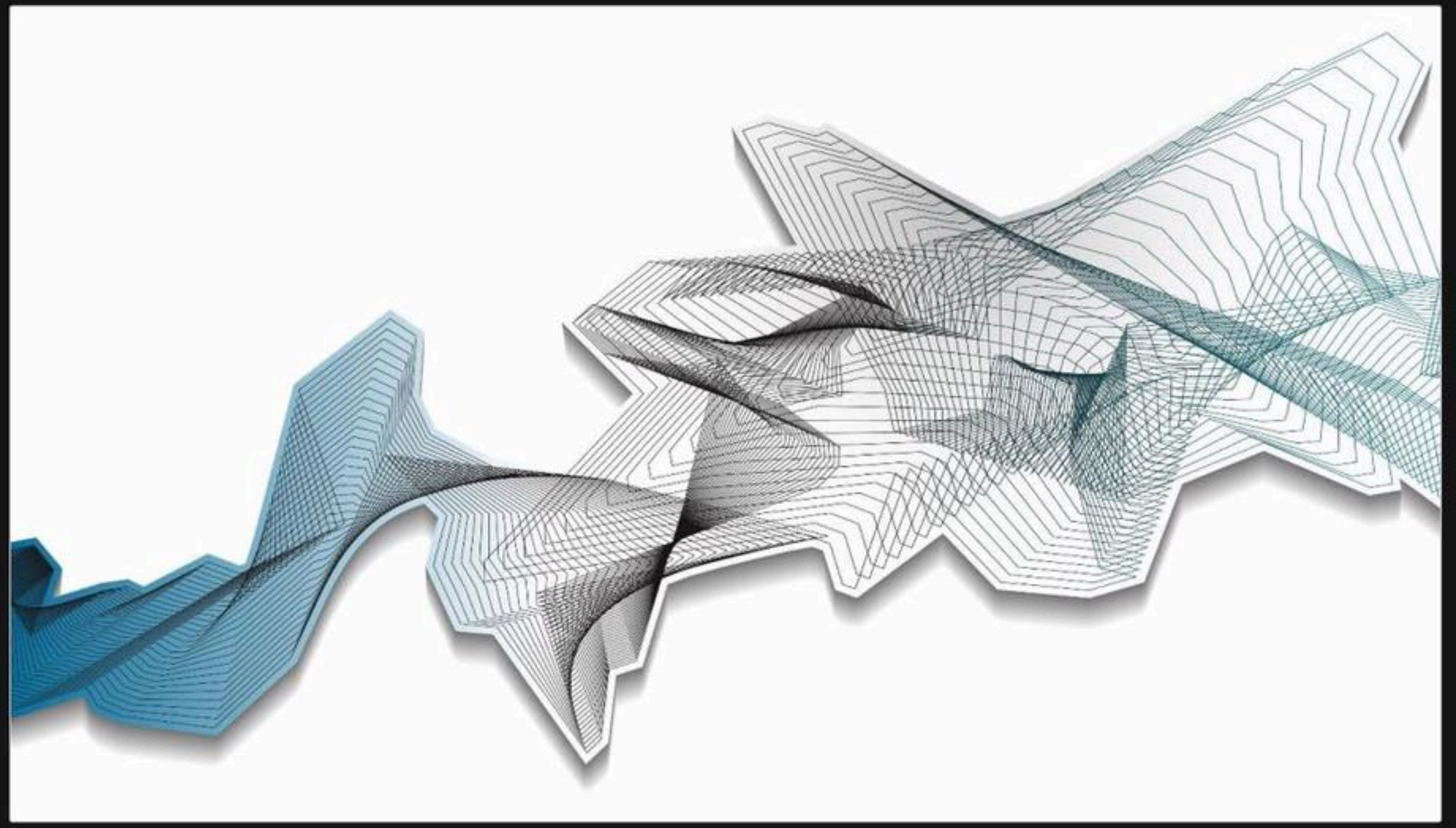




Manage code

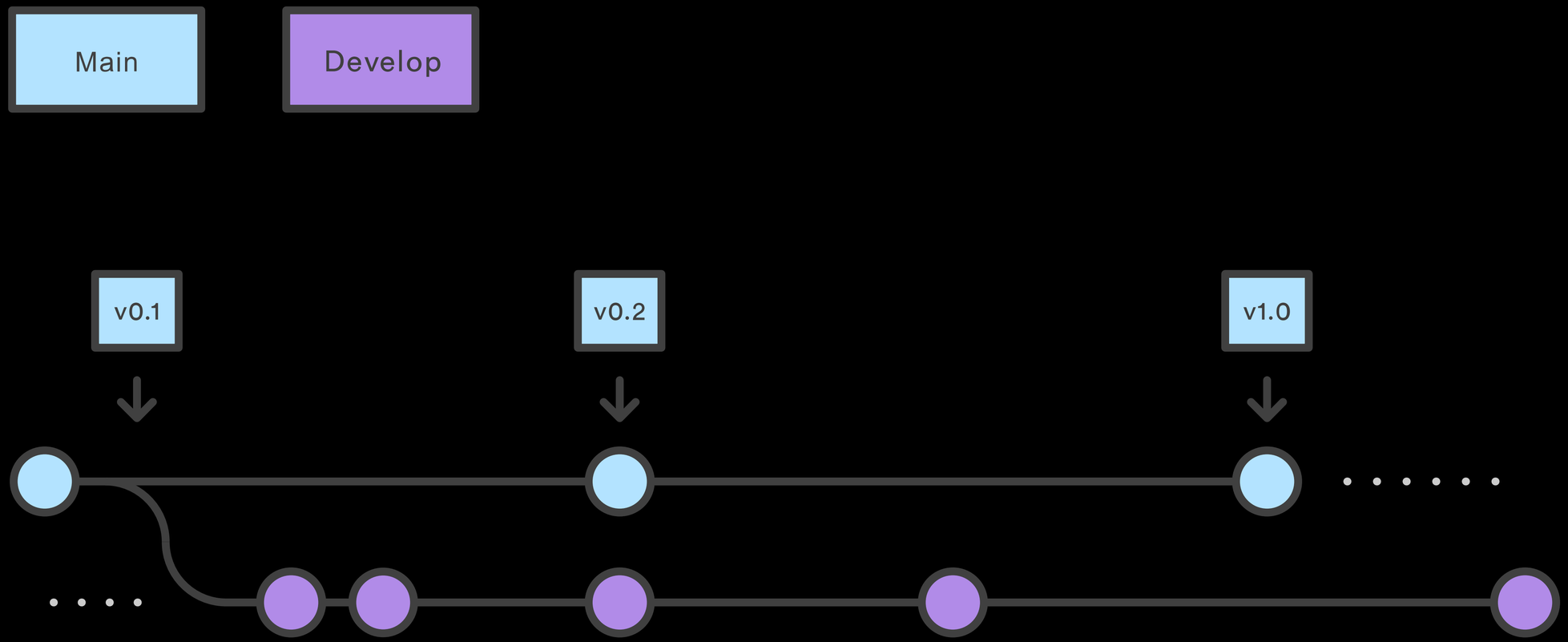
Git

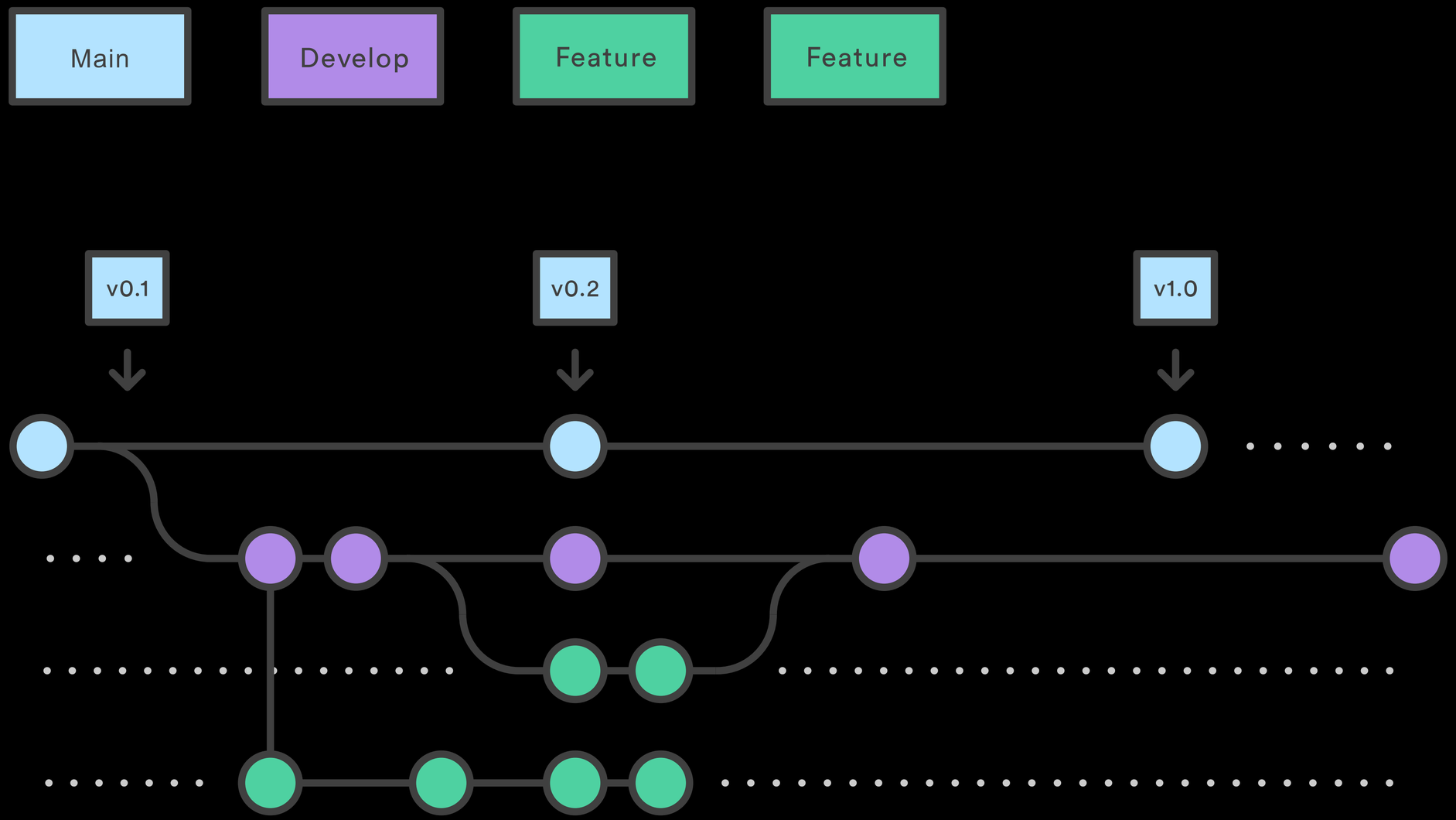
- The only sane way to work together on code
- https://learngitbranching.js.org/



Version Control

Git gets easier once you get the basic idea that branches are homeomorphic endofunctors mapping submanifolds of a Hilbert space.


Main
Develop
v0.1
v0.2
v1.0


Main
Develop
Feature
Feature
v0.1
v0.2
v1.0